ASSIGNMENT NO 7 SY-LAB MANUAL

Title:

Web Scraping and Analysis of Daily Vegetable Prices from AGMARKNET Portal

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Aim:

To collect daily vegetable price data from the government portal **http://agmarknet.gov.in/** (AGMARKNET) and perform basic data storage and analysis for an agri-analytics company.

Prerequisites:

Technical Requirements:

- Basic knowledge of Python programming
- Understanding of web scraping concepts
- Knowledge of data analysis using Pandas

Software Requirements:

- Python (3.x)
- Libraries: requests, beautifulsoup4, pandas, matplotlib
- Internet connection

System Requirements:

- Windows/Linux/Mac system
- Code editor (VS Code / PyCharm / Jupyter Notebook)

Objective:

1. To understand how government agricultural portals provide market price data.
2. To extract daily vegetable price data from the AGMARKNET portal.
3. To store collected data in structured format (CSV/Excel).
4. To perform basic analysis such as:
    - Average price calculation
    - Price comparison across markets
    - Trend visualization

Theory:

1. Introduction to AGMARKNET

AGMARKNET (Agricultural Marketing Information Network) is an initiative by the Government of India that provides agricultural commodity price data collected from various mandis (markets) across India. It is maintained under the Directorate of Marketing & Inspection.

The portal provides:

- Commodity-wise prices
- Market-wise prices
- State-wise agricultural data
- Daily minimum, maximum, and modal prices

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2. Web Scraping Concept

Web scraping is the process of extracting data from websites automatically using programming tools.

Steps Involved:

1. Send request to website
2. Receive HTML response
3. Parse HTML content
4. Extract required data
5. Store in structured format

Tools Used:

- requests → to fetch webpage
- BeautifulSoup → to parse HTML
- pandas → to organize and analyze data

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3. Data Processing

After extracting vegetable prices:

- Clean missing values
- Convert price columns into numeric format
- Store data in CSV/Excel
- Perform statistical analysis

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4. Applications in Agri-Analytics

Collected data can help:

- Farmers decide best market to sell produce
- Traders analyze price trends
- Companies forecast demand
- Government monitor inflation

Procedure:

1. Study the AGMARKNET website structure.
2. Identify commodity and date filters.
3. Write Python script to fetch webpage data.
4. Parse and extract vegetable price table.
5. Store data in CSV format.
6. Perform analysis (mean, max, min prices).
7. Visualize trends using graphs.

Expected Outcome:

- Successfully collected daily vegetable price dataset.
- Data stored in CSV/Excel format.
- Basic price trend analysis completed.

Conclusion:

In this experiment, we successfully studied and implemented a system to collect daily vegetable prices from the **AGMARKNET government portal**. The extracted data can be used by an agri-analytics company for price forecasting, trend analysis, and decision-making. Web scraping combined with data analytics provides a powerful solution for agricultural market intelligence.